

A PAIRED DIRICHLET SOURCE PACKAGE AND THE GRH TRANSFER SKELETON

WORKING DRAFT

ABSTRACT. This is a proof skeleton for the GRH transfer program suggested by the current RH manuscript and the GRH Dirichlet-tau research cycle. The paper separates three layers. First, it proves a completed-Hadamard source package for primitive nonprincipal Dirichlet pairs, in the normalization currently cleared by the verifier lane. Second, it records the local source-to-slot mechanism as a conditional theorem whose missing input is the paired unit-coordinate local normal form. Third, it states the formal GRH-family transfer theorem with every currently missing analytic or local step marked as a GAP. The zeta-specific source theorem is treated as an input from the RH manuscript, not reproved here.

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1. INTRODUCTION

The target of this draft is not to claim GRH. The target is to make the current proof shape editable: every claimed step is written as a theorem or a lemma, and every missing step is marked as a GAP with its current UV label.

The message is the following. The RH manuscript appears to contain a local phase, jet, whitening, and odd-projector package which is largely independent of zeta once a source scalar has been supplied. The first non-zeta source target is the primitive Dirichlet pair

$$\Xi_\chi(s) := \Lambda(s, \chi)\Lambda(s, \bar{\chi}), \quad \Phi_\chi^{\text{pair}}(s) := \frac{\Xi_\chi(2s-1)}{\Xi_\chi(2s)}.$$

For this object, the completed-zero source package can be written cleanly on regular compact boundary intervals. The remaining obstruction is not the source sign or a hidden factor of 2. It is the local normal-form question: does the paired local block use the same scalar as its zeroth value coordinate?

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1.1. What this draft proves. The unconditional theorem in this draft is the completed paired source package for primitive nonprincipal characters; see Theorem 4.4. It proves the sign-audited identity

$$q_\chi^{\text{pair}}(t) = \text{Re} \sum_{\rho \in Z(\Xi_\chi)} m_\rho \left(\frac{1}{1 + 2it - \rho} - \frac{1}{2it - \rho} \right)$$

on regular compact intervals, with the completed-Hadamard normalization

$$B_{\chi, \text{comp}}^{\text{pair}} = 0, \quad S_{\chi, \text{comp}}^{\text{pair}} = q_\chi^{\text{pair}}.$$

1.2. What this draft does not prove. This draft does not prove GRH, ERH, or a tau-family analogue. Those statements require the source-to-slot, odd/transverse, remainder, and final contradiction layers listed in Sections 5–7. They are recorded as GAPs rather than hidden in prose.

2. THE ZETA-SIDE INPUT

The zeta case is treated as special in this paper. The RH manuscript is the place where the zeta source theorem, local slot theorem, odd transverse package, and final contradiction are meant to be proved. This draft only records the interface required from that manuscript.

Hypothesis 2.1 (Zeta source interface). Let

$$\phi_\zeta(s) = \frac{\Lambda_\zeta(2s - 1)}{\Lambda_\zeta(2s)}.$$

On each regular compact boundary interval, the RH manuscript supplies a phase convention, a source derivative q_ζ , a background term B_ζ , and a positive strip-edge source scalar S_ζ such that

$$q_\zeta(t) = B_\zeta(t) + S_\zeta(t).$$

The source scalar is the same scalar that enters the RH local value slot.

GAP 2.2 (UV-019: zeta source theorem). The canonical zeta source theorem for q_ζ , B_ζ , and S_ζ is delegated to the RH paper. The GRH transfer draft should not reprove it and should not use it without the phase convention, compact summation, background naming, and multiplicity convention fixed there.

3. PRIMITIVE DIRICHLET COMPLETED FUNCTIONS

Let χ be a primitive nonprincipal Dirichlet character modulo r , and put

$$a_\chi := \frac{1 - \chi(-1)}{2}.$$

We use the completed L -function normalization

$$\Lambda(s, \chi) := \left(\frac{r}{\pi} \right)^{(s+a_\chi)/2} \Gamma\left(\frac{s+a_\chi}{2} \right) L(s, \chi).$$

Proposition 3.1 (Standard completed Dirichlet input). *For primitive nonprincipal χ , the completed function $\Lambda(s, \chi)$ is entire of order at most one and satisfies the standard primitive functional equation relating χ and $\bar{\chi}$. The paired product*

$$\Xi_\chi(s) := \Lambda(s, \chi) \Lambda(s, \bar{\chi})$$

is entire of order at most one. Its zeros carry analytic multiplicities, and the nonzero zeros satisfy

$$\sum_{\rho \neq 0} m_\rho |\rho|^{-2} < \infty.$$

Proof. The continuation, primitive functional equation, trivial-zero locations, and $L(1, \chi) \neq 0$ input are the standard Dirichlet L -function facts, for example in DLMF Section 25.15. The displayed completed normalization is obtained from the primitive functional equation by the parity choice $a_\chi = (1 - \chi(-1))/2$. The gamma poles are cancelled by the corresponding trivial zeros, so the completed primitive nonprincipal function is entire.

For this working draft, the order-one input is taken in the standard bounded form: combine the DLMF growth formulae for $L(s, \chi)$, the primitive functional equation, and Stirling's formula for Γ . This gives finite order at most one for $\Lambda(s, \chi)$, hence also for Ξ_χ . Hadamard factorization for entire functions of finite order then gives a genus-one product, analytic zero multiplicities, and the summability $\sum m_\rho |\rho|^{-2} < \infty$. A final publication draft should replace this paragraph by exact textbook page or theorem references. $\square$

GAP 3.2 (Bibliography upgrade for Proposition 3.1). The working citation stack is DLMF Section 25.15, DLMF Section 5.11 for Stirling, and a standard Hadamard factorization theorem such as Conway, Chapter XI, Section 3. A final draft should verify exact Davenport or Montgomery–Vaughan page and equation references.

4. THE COMPLETED PAIRED SOURCE PACKAGE

Definition 4.1 (Paired quotient). For primitive nonprincipal χ , define

$$\Phi_\chi^{\text{pair}}(s) := \frac{\Xi_\chi(2s-1)}{\Xi_\chi(2s)}.$$

Let $I \subset \mathbb{R}$ be compact and assume that

$$2it \notin Z(\Xi_\chi), \quad 1+2it \notin Z(\Xi_\chi)$$

for every $t \in I$. Such an interval is called regular.

Definition 4.2 (Boundary phase and source derivative). On a regular compact interval I , choose a real C^1 boundary phase $\Theta_\chi^{\text{pair}}$ by

$$\Phi_\chi^{\text{pair}}\left(\frac{1}{2} + it\right) = e^{-2i\Theta_\chi^{\text{pair}}(t)}.$$

Set

$$q_\chi^{\text{pair}}(t) := \Theta_\chi^{\text{pair}'}(t).$$

Lemma 4.3 (Sign bridge). *On a regular compact interval,*

$$q_\chi^{\text{pair}}(t) = -\frac{1}{2} \frac{\Phi_\chi^{\text{pair}'}}{\Phi_\chi^{\text{pair}}}\left(\frac{1}{2} + it\right).$$

Equivalently, if

$$D_\chi(t) := \frac{\Xi_\chi'}{\Xi_\chi}(1+2it) - \frac{\Xi_\chi'}{\Xi_\chi}(2it),$$

then $q_\chi^{\text{pair}}(t) = D_\chi(t)$. In particular $D_\chi(t)$ is real on the chosen boundary branch.

Proof. Differentiate

$$\Phi_\chi^{\text{pair}}\left(\frac{1}{2} + it\right) = e^{-2i\Theta_\chi^{\text{pair}}(t)}$$

with respect to t . Since $s = 1/2 + it$, the left side contributes

$$i \frac{\Phi_\chi^{\text{pair}'}}{\Phi_\chi^{\text{pair}}}\left(\frac{1}{2} + it\right),$$

while the right side contributes $-2i\Theta_\chi^{\text{pair}'}(t)$. Hence

$$\frac{\Phi_\chi^{\text{pair}'}}{\Phi_\chi^{\text{pair}}}\left(\frac{1}{2} + it\right) = -2q_\chi^{\text{pair}}(t).$$

The chain rule for $\Phi_\chi^{\text{pair}}(s) = \Xi_\chi(2s - 1)/\Xi_\chi(2s)$ gives

$$\frac{\Phi_\chi^{\text{pair}'}}{\Phi_\chi^{\text{pair}}}(s) = 2 \frac{\Xi'_\chi}{\Xi_\chi}(2s - 1) - 2 \frac{\Xi'_\chi}{\Xi_\chi}(2s).$$

Putting $s = 1/2 + it$ yields

$$-\frac{1}{2} \frac{\Phi_\chi^{\text{pair}'}}{\Phi_\chi^{\text{pair}}}\left(\frac{1}{2} + it\right) = \frac{\Xi'_\chi}{\Xi_\chi}(1 + 2it) - \frac{\Xi'_\chi}{\Xi_\chi}(2it) = D_\chi(t).$$

□

Theorem 4.4 (Completed paired source package). *Let χ be primitive nonprincipal, and let $I \subset \mathbb{R}$ be a regular compact interval. Then*

$$q_\chi^{\text{pair}}(t) = \text{Re} \sum_{\rho \in Z(\Xi_\chi)} m_\rho \left(\frac{1}{1 + 2it - \rho} - \frac{1}{2it - \rho} \right), \quad t \in I,$$

where the sum runs over distinct completed zeros of Ξ_χ , and m_ρ is analytic multiplicity in $\Xi_\chi = \Lambda(\cdot, \chi)\Lambda(\cdot, \bar{\chi})$. The series converges absolutely and uniformly on I , and the same is true after any fixed number of t -derivatives.

Thus, in completed-Hadamard normalization,

$$q_{\chi, \text{comp}}^{\text{pair}}(t) = B_{\chi, \text{comp}}^{\text{pair}}(t) + S_{\chi, \text{comp}}^{\text{pair}}(t), \quad B_{\chi, \text{comp}}^{\text{pair}}(t) = 0,$$

with

$$S_{\chi, \text{comp}}^{\text{pair}}(t) := \text{Re} \sum_{\rho \in Z(\Xi_\chi)} m_\rho \left(\frac{1}{1 + 2it - \rho} - \frac{1}{2it - \rho} \right).$$

Proof. By Proposition 3.1, Ξ_χ is entire of order at most one. On any simply connected region avoiding its zeros, the genus-one Hadamard product gives the logarithmic derivative in the form

$$\frac{\Xi'_\chi}{\Xi_\chi}(w) = b_\chi + \sum_{\rho \in Z(\Xi_\chi)} m_\rho \left(\frac{1}{w - \rho} + \frac{1}{\rho} \right),$$

with locally uniform convergence after the usual regularization. If a zero at $\rho = 0$ is admitted by a broader convention, it is separated as a finite factor. In the primitive nonprincipal setting used here, the boundary edge points in I avoid zeros by regularity.

Take the edge difference at $w = 1 + 2it$ and $w = 2it$. The constant b_χ cancels, and the regularizing $1/\rho$ terms cancel term by term. Hence

$$D_\chi(t) = \sum_{\rho \in Z(\Xi_\chi)} m_\rho \left(\frac{1}{1 + 2it - \rho} - \frac{1}{2it - \rho} \right).$$

For $t \in I$, regularity gives a positive distance from the two moving edge points to the zero set. For large $|\rho|$, each summand is

$$O_I(|\rho|^{-2}),$$

because it is the difference of two resolvents whose arguments differ by 1. The zero summability from Proposition 3.1 then gives absolute and uniform convergence on I . After k derivatives in t , the summand is a constant multiple of the difference of two $(k + 1)$ -st powers of the same resolvents, hence is $O_{I,k}(|\rho|^{-k-2})$. This proves uniform convergence after every fixed number of derivatives.

By Lemma 4.3, $q_\chi^{\text{pair}} = D_\chi$, and the boundary phase makes this quantity real. Keeping Re in the display records the scalar source identity. Since the completed-zero theorem already sums every completed zero of Ξ_χ with analytic multiplicity, there is no further completed background term. This is exactly $B_{\chi, \text{comp}}^{\text{pair}} = 0$. □

Remark 4.5 (Raw L -factor bookkeeping). The theorem uses completed zeros of Ξ_χ . In this normalization there is no additional gamma, trivial-zero, pole, conductor, or Hadamard-exponential background term. If one rewrites the same identity using raw zeros of $L(s, \chi)$ and $L(s, \bar{\chi})$, those raw background pieces must be placed in a separate raw bookkeeping term. They must not be added to Theorem 4.4.

GAP 4.6 (UV-016: final promotion bookkeeping). The mathematical wording of Theorem 4.4 follows the verifier-cleared staged source theorem. The remaining UV-016 work for a final paper is bibliographic polish and any decision about whether to keep the completed normalization $B_{\chi, \text{comp}}^{\text{pair}} = 0$ as the main theorem or add a separate raw-normalization corollary.

GAP 4.7 (UV-021: sign convention installed here). The sign convention is installed in Definition 4.2 and Lemma 4.3: fixed quotient, negative boundary exponent, $q = \Theta'$, and

$$q_\chi^{\text{pair}} = -\frac{1}{2}(\Phi_\chi^{\text{pair}'} / \Phi_\chi^{\text{pair}})(1/2 + it).$$

Older positive-exponent text should be treated as obsolete unless the quotient is inverted or q is redefined.

5. THE SOURCE-TO-SLOT LAYER

The source package is not yet a GRH theorem. It must feed the same local value channel used by the RH manuscript. The following theorem records the exact conditional mechanism.

Hypothesis 5.1 (Paired unit-coordinate local chart). For each $m \in I$, the corrected paired local block is represented near the background by a holomorphic map

$$\mathcal{F}_{\chi, m, \sigma}(a, \eta),$$

where a is the zeroth value coordinate

$$a = q_\chi^{\text{pair}}(m) - B_\chi^{\text{pair}}(m)$$

and η contains all non-value local coordinates: derivative, mixed-jet, curvature, background, cutoff, correction, multiplicity, and normalization data. In the completed normalization of Theorem 4.4,

$$a = S_{\chi, \text{comp}}^{\text{pair}}(m) = q_\chi^{\text{pair}}(m).$$

The pure value path is

$$(a, \eta) = (\alpha, \eta_0),$$

so all non-value coordinates are frozen.

Hypothesis 5.2 (Holomorphic whitening). The corrected paired same-point and mixed blocks are holomorphic in (a, η) near the background, the same-point blocks are positive and nondegenerate there, and the inverse-square-root whitening map is holomorphic.

Theorem 5.3 (Conditional paired source-to-slot realization). *Assume Theorem 4.4, Hypothesis 5.1, and Hypothesis 5.2. Define*

$$A_{\text{val}, \chi}^{\text{pair}}(m, \sigma) := \partial_a \mathcal{F}_{\chi, m, \sigma}(0, \eta_0)$$

and

$$R_\chi^{\text{pair}}(m, \sigma) := \mathcal{F}_{\chi, m, \sigma}(S_{\chi, \text{comp}}^{\text{pair}}(m), \eta_\chi(m)) - \mathcal{F}_{\chi, m, \sigma}(0, \eta_0) - S_{\chi, \text{comp}}^{\text{pair}}(m) A_{\text{val}, \chi}^{\text{pair}}(m, \sigma).$$

Then

$$\Delta_\chi^{\text{pair}}(m, \sigma) = S_{\chi, \text{comp}}^{\text{pair}}(m) A_{\text{val}, \chi}^{\text{pair}}(m, \sigma) + R_\chi^{\text{pair}}(m, \sigma).$$

Moreover, along the pure value path the remainder has zero first derivative after subtracting the value term:

$$\left. \frac{\partial}{\partial \alpha} \left(\mathcal{F}_{\chi, m, \sigma}(\alpha, \eta_0) - \mathcal{F}_{\chi, m, \sigma}(0, \eta_0) - \alpha A_{\text{val}, \chi}^{\text{pair}}(m, \sigma) \right) \right|_{\alpha=0} = 0.$$

Proof. This is the local algebra of a holomorphic coordinate chart. By Hypothesis 5.1, the source scalar from Theorem 4.4 is the value coordinate a . By Hypothesis 5.2, the corrected whitened block is holomorphic in that coordinate and in the frozen non-value coordinates. The pure value derivative at the background is therefore

$$\partial_a \mathcal{F}_{\chi, m, \sigma}(0, \eta_0) = A_{\text{val}, \chi}^{\text{pair}}(m, \sigma).$$

The displayed definition of R_{χ}^{pair} is then an identity. The last display follows by differentiating the same expression along $(a, \eta) = (\alpha, \eta_0)$ at $\alpha = 0$. The proof is non-tautological only because the value coordinate was fixed before R_{χ}^{pair} was defined. $\square$

Corollary 5.4 (No hidden scalar if the tested value derivative is nonzero). *Assume the hypotheses of Theorem 5.3 and suppose the tested value derivative $A_{\text{val}, \chi}^{\text{pair}}(m, \sigma)$ is nonzero in the slot under consideration. A replacement*

$$c_{\chi}(m) S_{\chi, \text{comp}}^{\text{pair}}(m)$$

can satisfy the same pure value derivative test only if $c_{\chi}(m) = 1$.

Proof. If the coordinate were

$$a = c_{\chi}(m) S_{\chi, \text{comp}}^{\text{pair}}(m) + O(S^2),$$

then the first derivative of the pure value path with respect to the source scalar would be

$$c_{\chi}(m) A_{\text{val}, \chi}^{\text{pair}}(m, \sigma).$$

Equality with the unit coefficient derivative from Theorem 5.3 forces $c_{\chi}(m) = 1$ when the tested derivative is nonzero. $\square$

GAP 5.5 (UV-017: paired unit-coordinate normal form). The missing proof is Hypothesis 5.1. One must construct the paired corrected local blocks from the same boundary phase $\Theta_{\chi}^{\text{pair}}$ and prove that their zeroth coordinate is exactly

$$q_{\chi}^{\text{pair}}(m) - B_{\chi}^{\text{pair}}(m),$$

not $c_{\chi}(m)(q - B)$ and not an upstairs-only surrogate. The paired holomorphic-whitening hypotheses and the scalar readout normalization also belong to UV-017 unless the theorem is deliberately stated with them as assumptions.

6. ODD/TRANSVERSE AND REMAINDER LAYERS

The RH manuscript also uses a corrected odd transverse scalar, boundary control, and an N -point odd-moment projector. The paired Dirichlet source package does not supply these ingredients.

Hypothesis 6.1 (Odd/transverse package). For the family under consideration, the corrected local source-to-slot block produces an odd holomorphic transverse germ with the same kind of boundary control used in the RH manuscript. The discrete N -point odd projector annihilates the lower odd jets and leaves only the controlled high-order transverse tail.

Hypothesis 6.2 (Remainder dominance). The local remainder R_{χ}^{pair} is smaller than the leading value-channel term in the calibrated regime required by the RH contradiction argument.

GAP 6.3 (UV-018: corrected odd/transverse realization). The corrected odd transverse scalar, boundary estimate, and N -point projector input have not been constructed for primitive Dirichlet pairs. This is downstream from UV-017 and should not be folded into the source theorem.

GAP 6.4 (Remainder dominance). The current GRH team ledger does not yet close the paired remainder dominance needed to compare the leading value term with the transverse/projected error terms. This is a separate analytic estimate, not a source-normalization problem.

7. FORMAL TRANSFER THEOREM

This section states the paper's long target in a form that can be revised as gaps close.

Hypothesis 7.1 (RH local contradiction template). The RH manuscript supplies the following local contradiction template. If a family has:

- (1) a source scalar occupying the exact local value slot;
- (2) holomorphic whitening and same-point positivity;
- (3) odd/transverse realization and boundary control;
- (4) remainder dominance in the calibrated regime;
- (5) the same off-line toy anomaly comparison;

then an off-line zero in that family produces an incompatible transverse local configuration.

Theorem 7.2 (Conditional primitive Dirichlet GRH transfer). *Let χ be primitive nonprincipal. Assume the completed paired source package of Theorem 4.4, the paired source-to-slot realization of Theorem 5.3, the odd/transverse package Hypothesis 6.1, the remainder dominance Hypothesis 6.2, and the RH local contradiction template Hypothesis 7.1. Then the paired local model admits no off-line zero configuration for $L(s, \chi)$ and $L(s, \bar{\chi})$ in the scope covered by those hypotheses.*

Proof. Theorem 4.4 supplies the completed source scalar with the correct sign and normalization. Theorem 5.3 identifies that scalar with the local value slot. Hypothesis 6.1 supplies the odd transverse channel and the finite projector. Hypothesis 6.2 makes the residual terms subordinate to the leading calibrated comparison. Hypothesis 7.1 then applies the RH manuscript's local contradiction argument to the paired Dirichlet local model. Therefore the off-line configuration is excluded inside the stated scope. $\square$

GAP 7.3 (Not yet a proof of GRH). Theorem 7.2 is conditional. It becomes a GRH-family theorem only after UV-017, UV-018, remainder dominance, and the RH contradiction template are promoted without assumptions.

8. TAU AND OTHER FAMILIES

The Ramanujan tau L -function remains a useful stress test because it is self-dual, but the current source and local realization package is weaker than the paired Dirichlet package.

GAP 8.1 (UV-020: tau compact source and localization package). For tau, one still needs a compact source decomposition, microscopic denominator and holomorphy control, the correct block scaling hierarchy, whitening gain and boundary estimates, and growth bookkeeping. This draft does not attempt to prove those statements.

9. CURRENT PROOF LEDGER

- Completed paired source theorem: proved for the working draft in Theorem 4.4; final citation polish is GAP 3.2.
- Paired sign convention: installed in Lemma 4.3; GAP 4.7 records the obsolete convention risk.
- Zeta source theorem: delegated to the RH manuscript via Hypothesis 2.1 and GAP 2.2.
- Source-to-slot realization: conditional in Theorem 5.3; the missing local normal form is GAP 5.5.
- Odd/transverse package: missing, as recorded by Hypothesis 6.1 and GAP 6.3.
- Remainder dominance: missing, as recorded by Hypothesis 6.2 and GAP 6.4.
- GRH-family conclusion: conditional only; see Theorem 7.2 and GAP 7.3.
- Tau: exploratory; see GAP 8.1.

10. PROVENANCE

This skeleton is based on the current GRH team directory under `grh/teams/`, with timestamp 20260423-050630 and slug `research-team-grh-dirichlet-tau`. The core source theorem is drawn from `paper-updates.md`, the compact regularization reports, the background-multiplicity reports, and the Sartre verifier reports in the same team directory. The UV-017 reduction is drawn from Noether's coefficient-freeze report and its `coefficient_freeze_reduction.md` note. The live UV references are UV-016 through UV-021 in `uv.md`.

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- [3] J. B. Conway, *Functions of One Complex Variable I*, second edition, Springer, Chapter XI, Section 3.